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Definicion de variables

```
TPP <- rep(c(1, 2, 3), each = 3)

Chitosan <- rep(c(2, 3, 4), 3)

######################

PS <- c(106, 161, 182,
        145, 172, 246,
        158, 201, 321)

EE <- c(71.3, 58.2, 49.4,
        83.2, 64.5, 53.8,
        92.7, 87.1, 75.6)

Release <- c(77.2, 68.3, 57.9,
             85.8, 73.1, 62.6,
             97.1, 92.2, 81.1)
######################
```

Modelo de superficie de respuesta

```
library(rsm)
```

PS

```
Modelo_PS <- rsm(PS ~ FO(TPP, Chitosan) + TWI(TPP, Chitosan))
knitr::kable(summary(Modelo_PS)$lof[1:3,])
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
FO(TPP, Chitosan)	2	28160.167	14080.0833	42.318539	0.0007349
TWI(TPP, Chitosan)	1	1892.250	1892.2500	5.687271	0.0627906
Residuals	5	1663.583	332.7167	NA	NA

EE

```
Modelo_EE <- rsm(EE ~ FO(TPP, Chitosan) + TWI(TPP, Chitosan))
knitr::kable(summary(Modelo_EE)$lof[1:3,])
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
FO(TPP, Chitosan)	2	1755.1350	877.56750	41.1434767	0.0007853
TWI(TPP, Chitosan)	1	5.7600	5.76000	0.2700492	0.6254764
Residuals	5	106.6472	21.32944	NA	NA

Release

```
Modelo_Release <- rsm(Release ~ FO(TPP, Chitosan) + TWI(TPP, Chitosan))
knitr::kable(summary(Modelo_Release)$lof[1:3,])
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
FO(TPP, Chitosan)	2	1318.54167	659.27083	46.7183249	0.0005815
TWI(TPP, Chitosan)	1	2.72250	2.72250	0.1929262	0.6788254
Residuals	5	70.55806	14.11161	NA	NA

Pareto

PS

```r
anova((lm(PS ~ TPP + Chitosan + TPP*Chitosan)))
```

```
## Analysis of Variance Table
##
## Response: PS
##              Df  Sum Sq Mean Sq F value    Pr(>F)
## TPP           1  8893.5  8893.5 26.7300 0.0035546 **
## Chitosan      1 19266.7 19266.7 57.9071 0.0006227 ***
## TPP:Chitosan  1  1892.2  1892.2  5.6873 0.0627906 .
## Residuals     5  1663.6   332.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

EE

```r
anova((lm(EE ~ TPP + Chitosan + TPP*Chitosan)))
```

```
## Analysis of Variance Table
##
## Response: EE
##              Df Sum Sq Mean Sq F value   Pr(>F)
## TPP           1 975.37  975.37  45.729 0.001074 **
## Chitosan      1 779.76  779.76  36.558 0.001784 **
## TPP:Chitosan  1   5.76    5.76   0.270 0.625476
## Residuals     5 106.65   21.33
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Release

```r
anova((lm(Release ~ TPP + Chitosan + TPP*Chitosan)))
```

```
## Analysis of Variance Table
##
## Response: Release
##              Df Sum Sq Mean Sq F value    Pr(>F)
## TPP           1 748.17  748.17 53.0178 0.0007643 ***
## Chitosan      1 570.38  570.38 40.4188 0.0014227 **
## TPP:Chitosan  1   2.72    2.72  0.1929 0.6788254
## Residuals     5  70.56   14.11
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Superficie

```r
library(rsm)

library(plot3D)
```

OPTIMOS

```r
optimos <- function(modelo, x, y){
  print(modelo$coefficients[[1]] + modelo$coefficients[[2]]*x + modelo$coefficients[[3]]*y + modelo$coe
}


optimos(Modelo_PS, -0.605364, 1.229885)
```

```
## [1] 87.69349
```

```r
optimos(Modelo_EE, 11.500, -7.625)
```

```
## [1] 191.7694
```

```r
optimos(Modelo_Release, 0.4125, -0.4125)
```

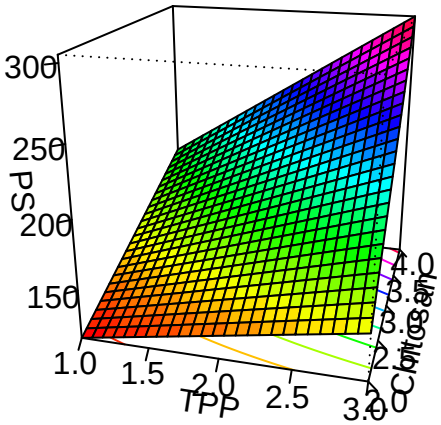
```
## [1] 97.26966
```

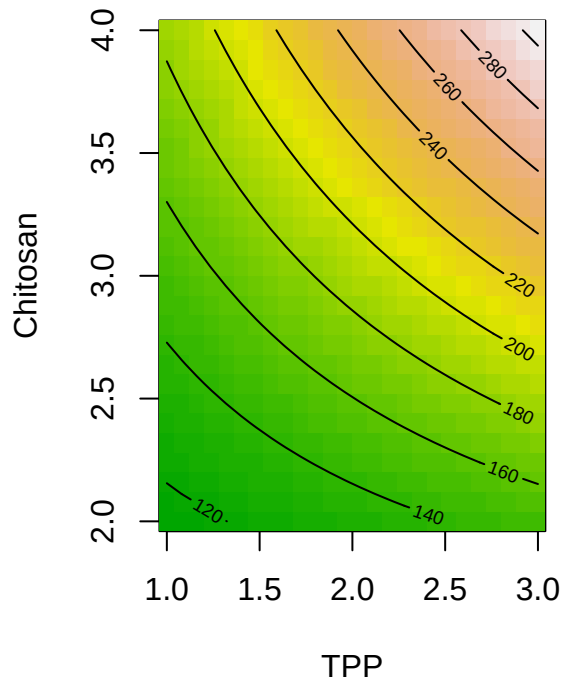
Superficie de respuesta

PS

```r
par(mfrow = c(1,2))
persp(Modelo_PS, Chitosan ~ TPP,
      zlab = "PS",
      contours = list(z = "bottom", col = "colors"), # posicion y color
      at = c(summary(Modelo_PS$canonical$xs)),
      theta = 15, # coordenadas graficas
      phi = 20)

contour(Modelo_PS, ~ TPP + Chitosan, image = TRUE)
```

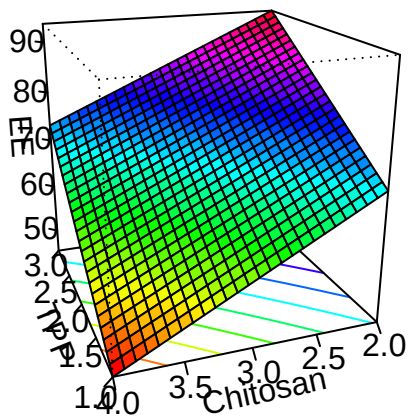


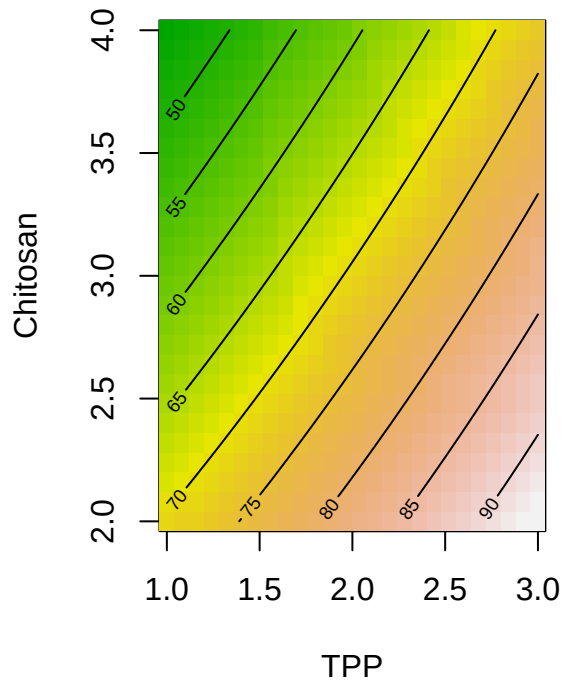


EE

```
par(mfrow = c(1,2))
persp(Modelo_EE, Chitosan ~ TPP,
      zlab = "EE",
      contours = list(z = "bottom", col = "colors"), # posicion y color
      at = c(summary(Modelo_PS$canonical$xs)),
      theta = 250, # coordenadas graficas
      phi = 20)

contour(Modelo_EE, ~ TPP + Chitosan, image = TRUE)
```

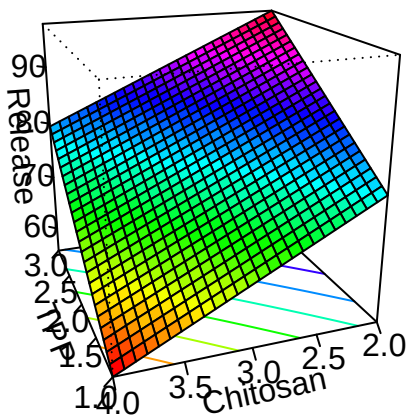


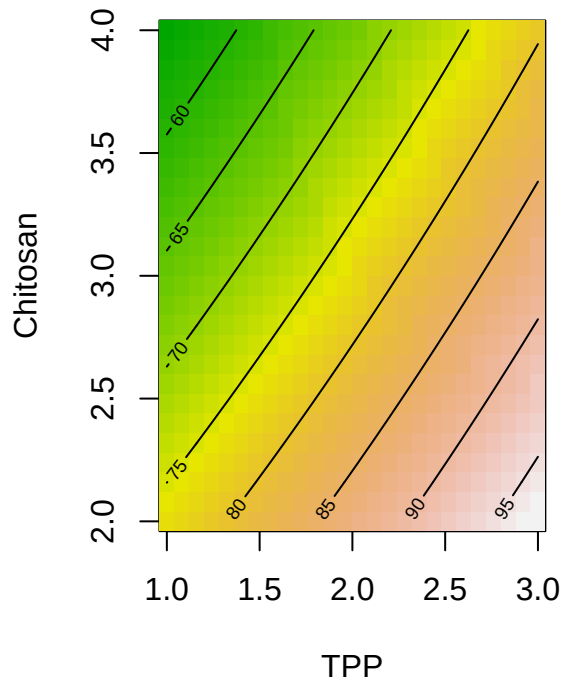


Release

```
par(mfrow = c(1,2))
persp(Modelo_Release, Chitosan ~ TPP,
      zlab = "Release",
      contours = list(z = "bottom", col = "colors"), # posicion y color
      at = c(summary(Modelo_PS$canonical$xs)),
      theta = 250, # coordenadas graficas
      phi = 20)

contour(Modelo_Release, ~ TPP + Chitosan, image = TRUE)
```






```
# col = c("#0000FF", "#006CFF", "#00FF00", "#8FFF00",
#         "#FFFF00", "#FFCC00", "#FF9C00", "#FF5A04", "#FF0000")
```